1.9.25

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Question 1.9.25

If the point $P(x, y)$ is equidistant from

$$A(a+b, b-a) \quad \text{and} \quad B(a-b, a+b),$$

prove that

$$bx = ay.$$

Given Data

$$A = \begin{pmatrix} a+b \\ b-a \end{pmatrix}, \quad B = \begin{pmatrix} a-b \\ a+b \end{pmatrix}, \quad P = \begin{pmatrix} x \\ y \end{pmatrix} \tag{1}$$

Equidistant Condition

$$\|P - A\|^2 = \|P - B\|^2 \tag{2}$$

$$(P - A)^T (P - A) = (P - B)^T (P - B) \tag{3}$$

$$P^T P - 2A^T P + A^T A = P^T P - 2B^T P + B^T B \tag{4}$$

Simplification

$$2(B-A)^T P = B^T B - A^T A \tag{5}$$

$$B - A = \begin{pmatrix} -2b \\ 2a \end{pmatrix} = 2\begin{pmatrix} -b \\ a \end{pmatrix} \tag{6}$$

$$A^T A = (a+b)^2 + (b-a)^2 = 2a^2 + 2b^2 \tag{7}$$

$$B^T B = (a-b)^2 + (a+b)^2 = 2a^2 + 2b^2 \tag{8}$$

$$B^T B - A^T A = 0 \tag{9}$$

$$(B-A)^T P = 0 \tag{10}$$

Final Result

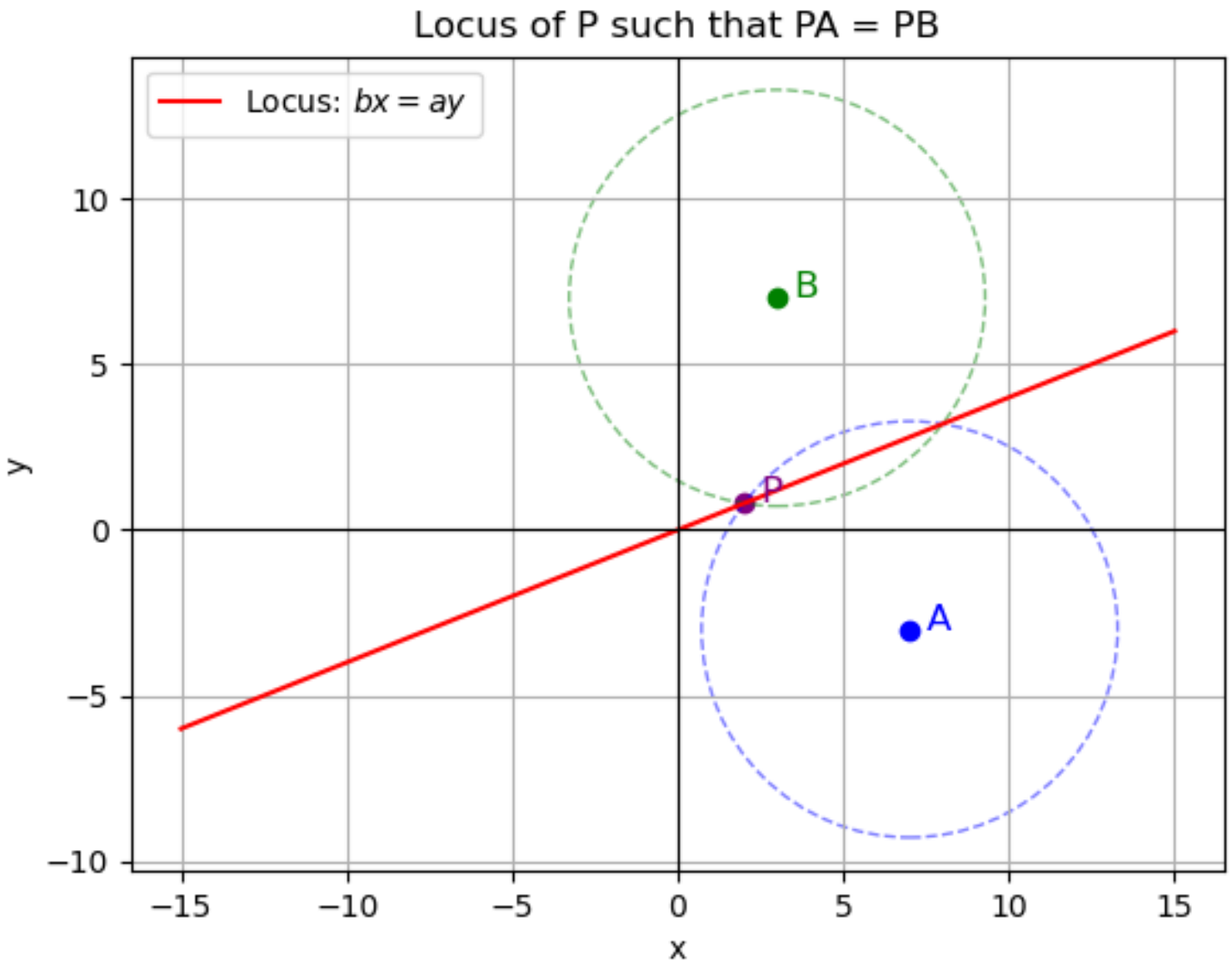
$$\begin{pmatrix} -2b & 2a \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \tag{11}$$

$$-2bx + 2ay = 0 \quad \Rightarrow \quad \boxed{bx = ay} \tag{12}$$

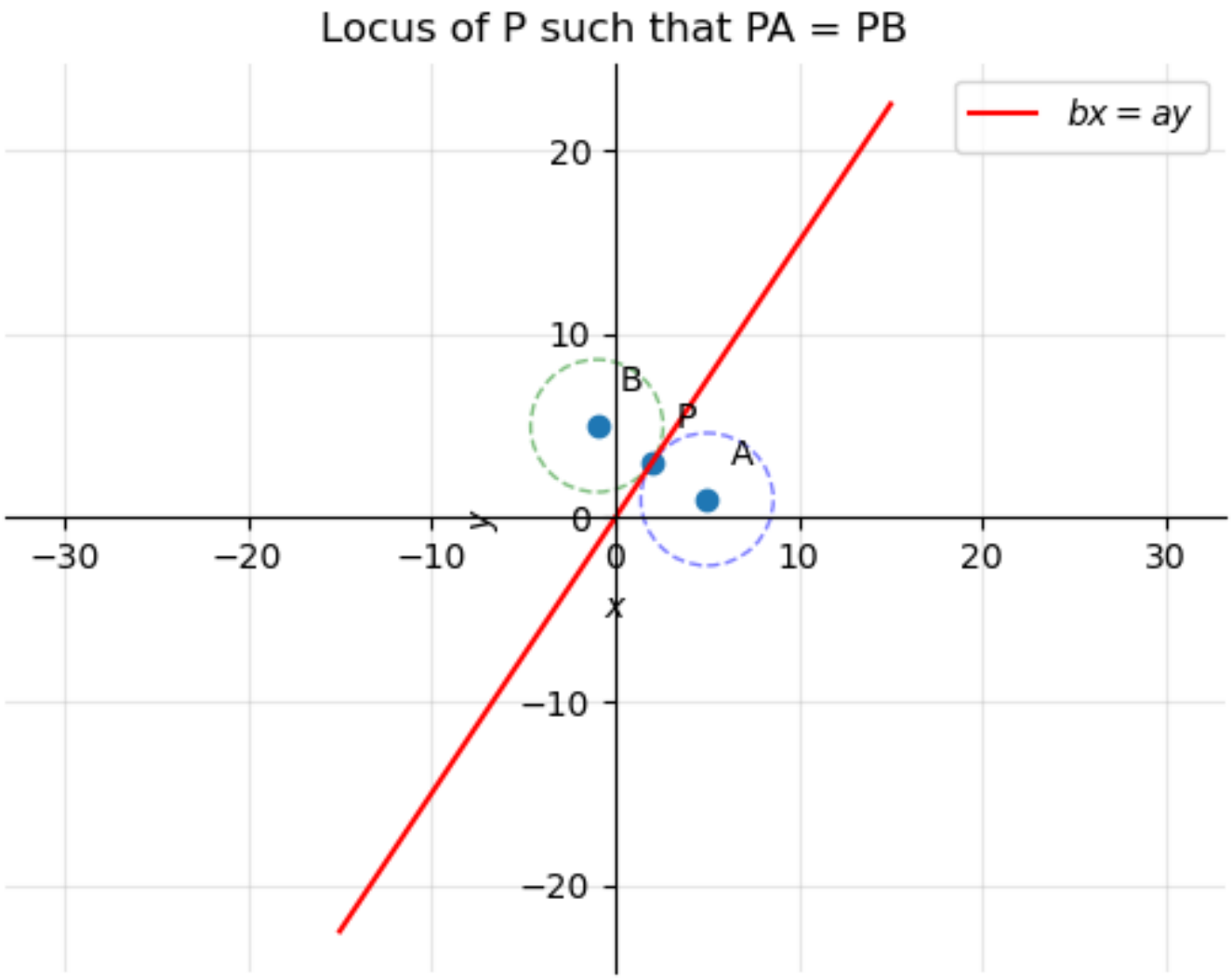
Thus, the locus of P is the line

$$n^T x = 0, \quad n = \begin{pmatrix} -b \\ a \end{pmatrix}.$$

Plot by shared output



Direct python code plot



C Code

```c
#include <stdio.h>

// Function to check if bx = ay
// Returns 1 if true, 0 otherwise
int check_equation(double a, double b, double x, double y) {
    double lhs = b * x;
    double rhs = a * y;
    if ( (lhs - rhs) < 1e-6 && (rhs - lhs) < 1e-6 ) {
        return 1; // condition satisfied
    }
    return 0; // not satisfied
}
```

Python Code through shared output

```
import ctypes \\
import numpy as np
import matplotlib.pyplot as plt

# Parameters
a, b = 5, 2
A = np.array([a+b, b-a])
B = np.array([a-b, a+b])

# Example point P satisfying bx = ay
P = np.array([2.0, (b/a)*2.0]) # pick x=2, y=(b/a)x

# Line bx = ay -> y = (b/a)x
x_vals = np.linspace(-15, 15, 400)
y_vals = (b/a) * x_vals

# Plot line (locus)
plt.plot(x_vals, y_vals, 'r-', label=r'Locus: $bx = ay$')
```

Python Code through shared output

```
# Plot points
plt.scatter(*A, color='blue')
plt.scatter(*B, color='green')
plt.scatter(*P, color='purple')


\begin{frame}[fragile]\frametitle{Python Code through shared
    output}
\begin{lstlisting}
# Annotate points
plt.text(A[0]+0.5, A[1], 'A', fontsize=12, color='blue')
plt.text(B[0]+0.5, B[1], 'B', fontsize=12, color='green')
plt.text(P[0]+0.5, P[1], 'P', fontsize=12, color='purple')

# Circles showing equidistance
d_PA = np.linalg.norm(P - A)
circleA = plt.Circle(A, d_PA, color='blue', fill=False, linestyle
    ='--', alpha=0.5)
```

Python Code through shared output

```
circleB = plt.Circle(B, d_PA, color='green', fill=False,
    linestyle='--', alpha=0.5)
plt.gca().add_patch(circleA)
plt.gca().add_patch(circleB)

# Axes
plt.axhline(0, color='black', linewidth=0.8)
plt.axvline(0, color='black', linewidth=0.8)
plt.xlabel(x)
plt.ylabel(y)
plt.title(Locus of P such that PA = PB)
plt.legend()
plt.axis(equal)
plt.grid(True)
plt.show()
```

Direct Python code

```
import numpy as np
import numpy.linalg as LA
import matplotlib.pyplot as plt

# local imports
from line.funcs import *
from conics.funcs import circ_gen

# Given values
a, b = 2, 3
A = np.array([[a+b, b-a]]).reshape(-1,1) # (5,-3)
B = np.array([[a-b, a+b]]).reshape(-1,1) # (-1,7)

# Choose a point P on line bx = ay
```

Direct Python code

```
xP = 2
yP = (b*xP)/a
P = np.array(([xP, yP])).reshape(-1,1)

# Distances (radius of circles)
r = LA.norm(P - A)

# Generate locus line bx = ay
xx = np.linspace(-15,15,400)
yy = (b*xx)/a

# Plot locus line
plt.plot(xx, yy, 'r', label='$bx=ay$')

# Plot A, B, P
tri_coords = np.block([[A,B,P]])
plt.scatter(tri_coords[0,:], tri_coords[1,:])

# Labels
```

Direct Python code

```
vert_labels = ['A','B','P']
for i, txt in enumerate(vert_labels):
    plt.annotate(txt,
                 (tri_coords[0,i], tri_coords[1,i]),
                 textcoords=offset points,
                 xytext=(10,10),
                 ha='center')



# Circles centered at A and B with radius r
circleA = plt.Circle(A.flatten(), r, color='blue', fill=False,
    linestyle='--', alpha=0.5)
circleB = plt.Circle(B.flatten(), r, color='green', fill=False,
    linestyle='--', alpha=0.5)

ax = plt.gca() # <-- no plt.subplots()
ax.add_patch(circleA)
ax.add_patch(circleB)
```

Direct Python code

```
# Axes through origin
ax.spines['top'].set_color('none')
ax.spines['right'].set_color('none')
ax.spines['left'].set_position('zero')
ax.spines['bottom'].set_position('zero')

plt.xlabel('$x$')
plt.ylabel('$y$')
plt.title(Locus of P such that PA = PB)
plt.legend()
plt.grid(True, alpha=0.3)
plt.axis('equal')

#plt.savefig(locus.pdf) # if using termux
plt.show()
```